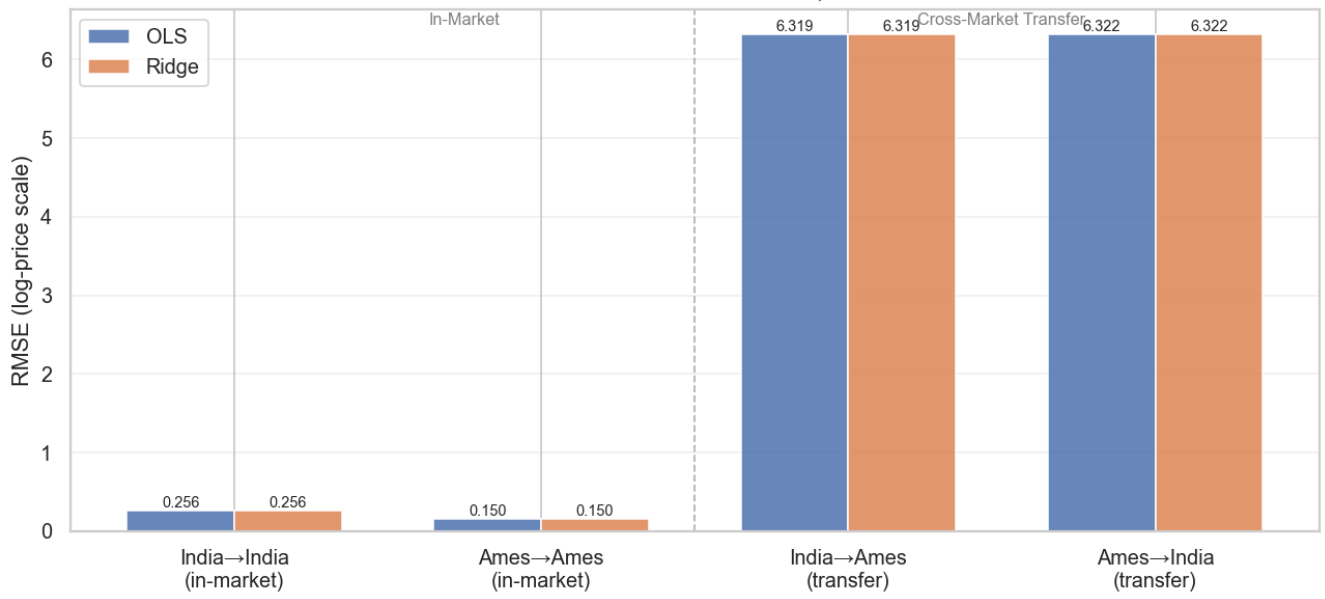
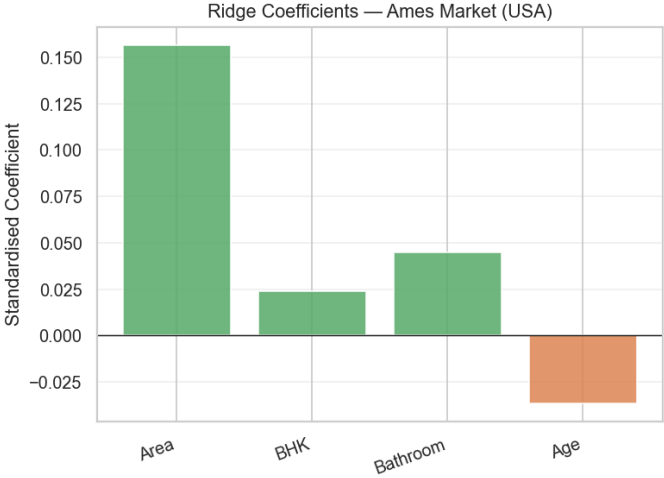
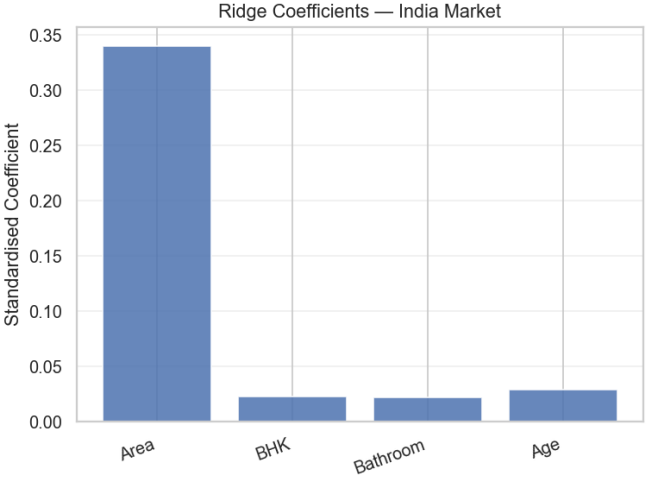


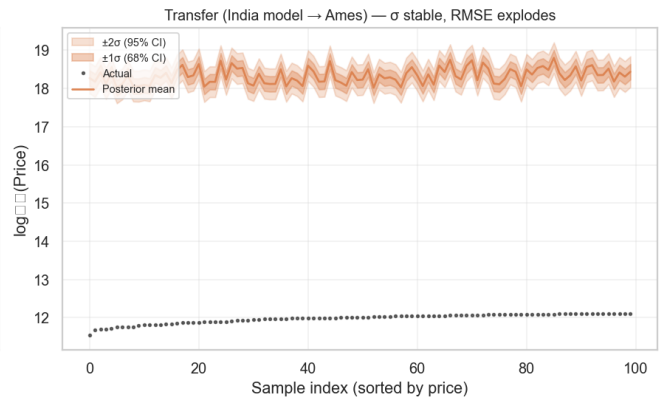
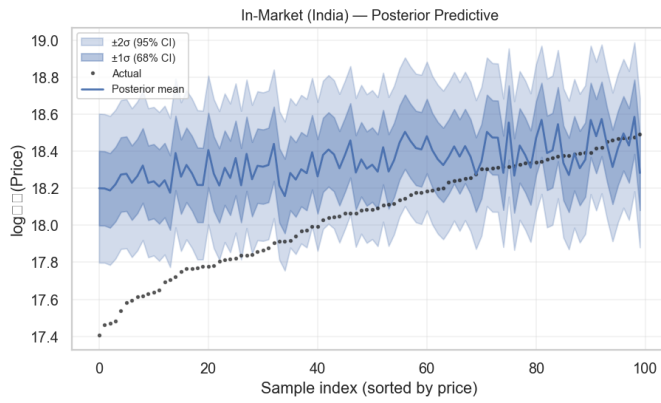
Phase 3 — Cross-Market Transfer: RMSE Comparison
Train on one market's universal features, test on the other



Universal Feature Weights: India vs Ames
Cosine Similarity = 0.9292 (1.0 = identical pricing logic)

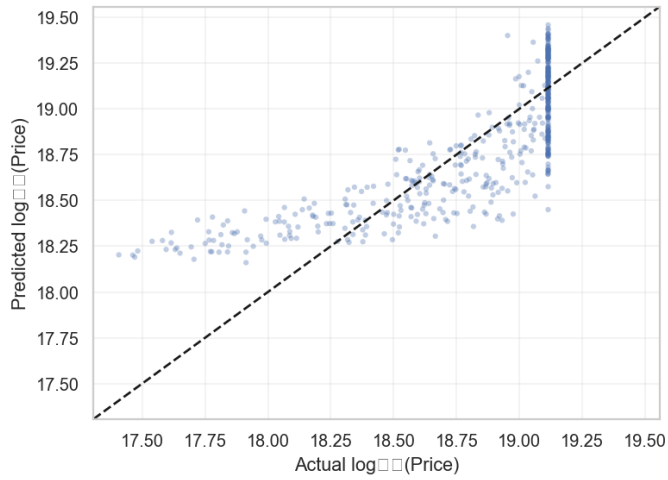


Bayesian LR — Posterior Predictive Intervals
Conjugate-Gaussian σ stays flat across markets; a market-specific
scale prior (Ch. 9.4) would widen intervals on transfer

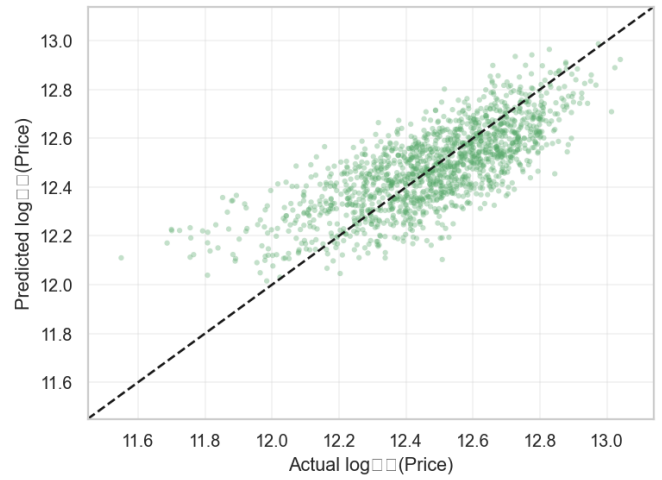


Phase 3 — Predicted vs Actual Across All Four Experiments
Transfer scatter is wider / more biased than in-market

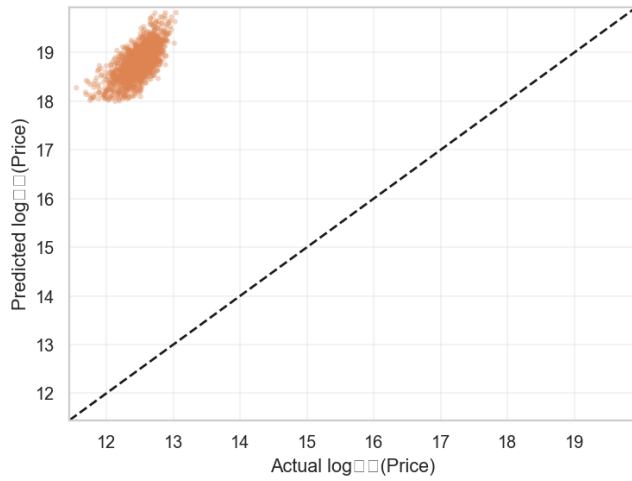
India→India (In-Market)
RMSE = 0.2543



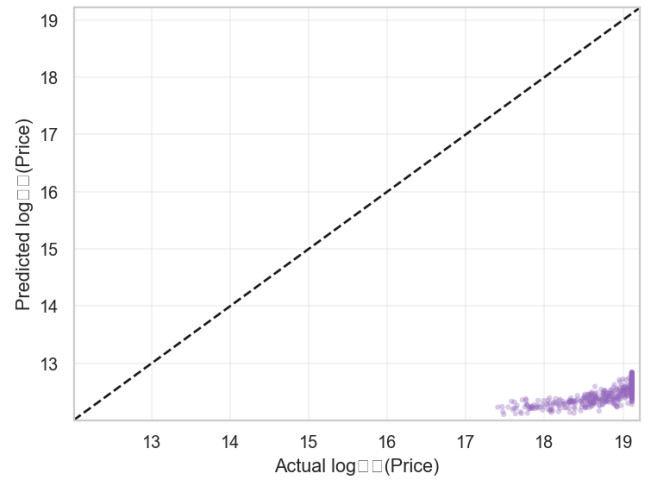
Ames→Ames (In-Market)
RMSE = 0.1500



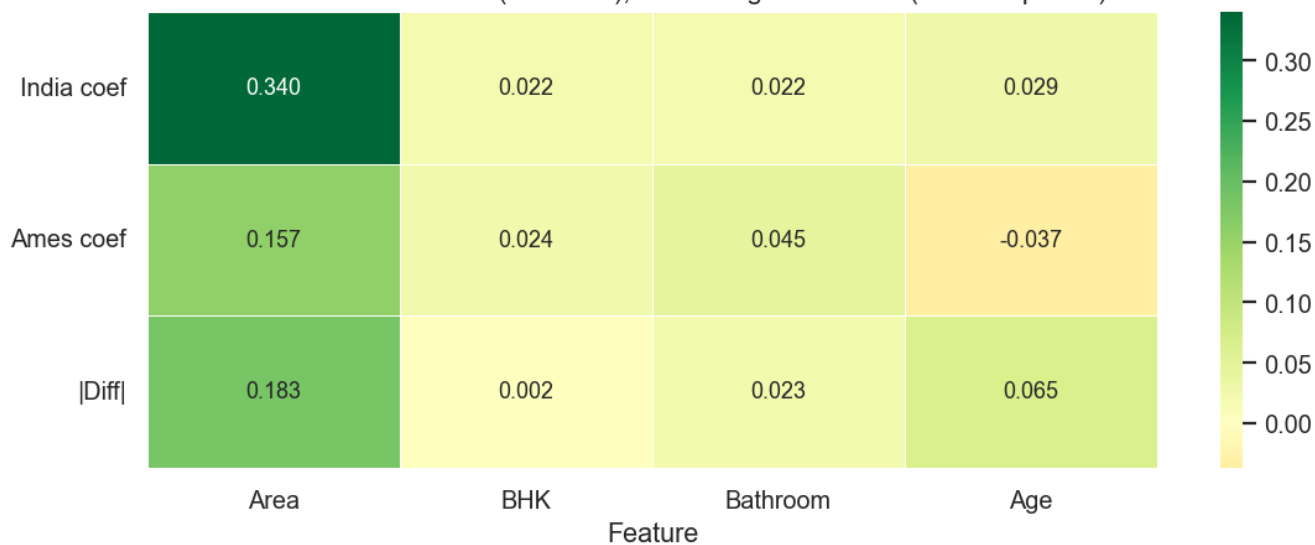
India→Ames (Transfer)
RMSE = 6.3192



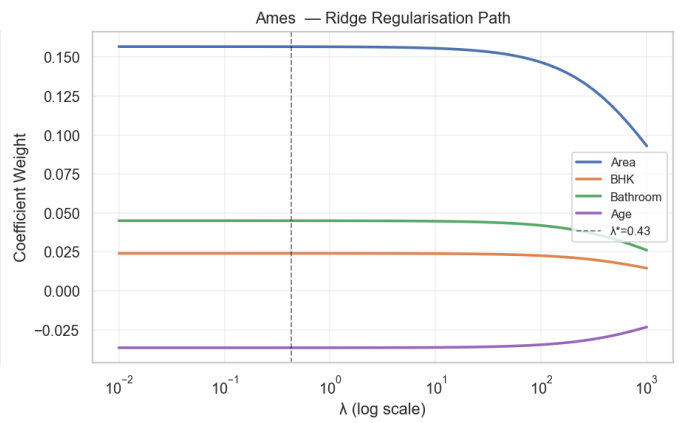
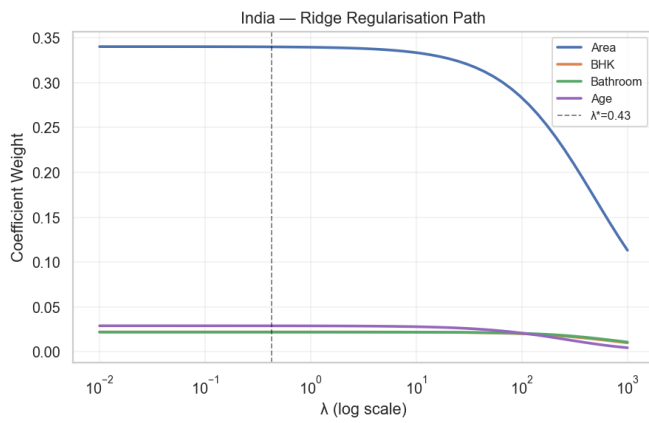
Ames→India (Transfer)
RMSE = 6.3224



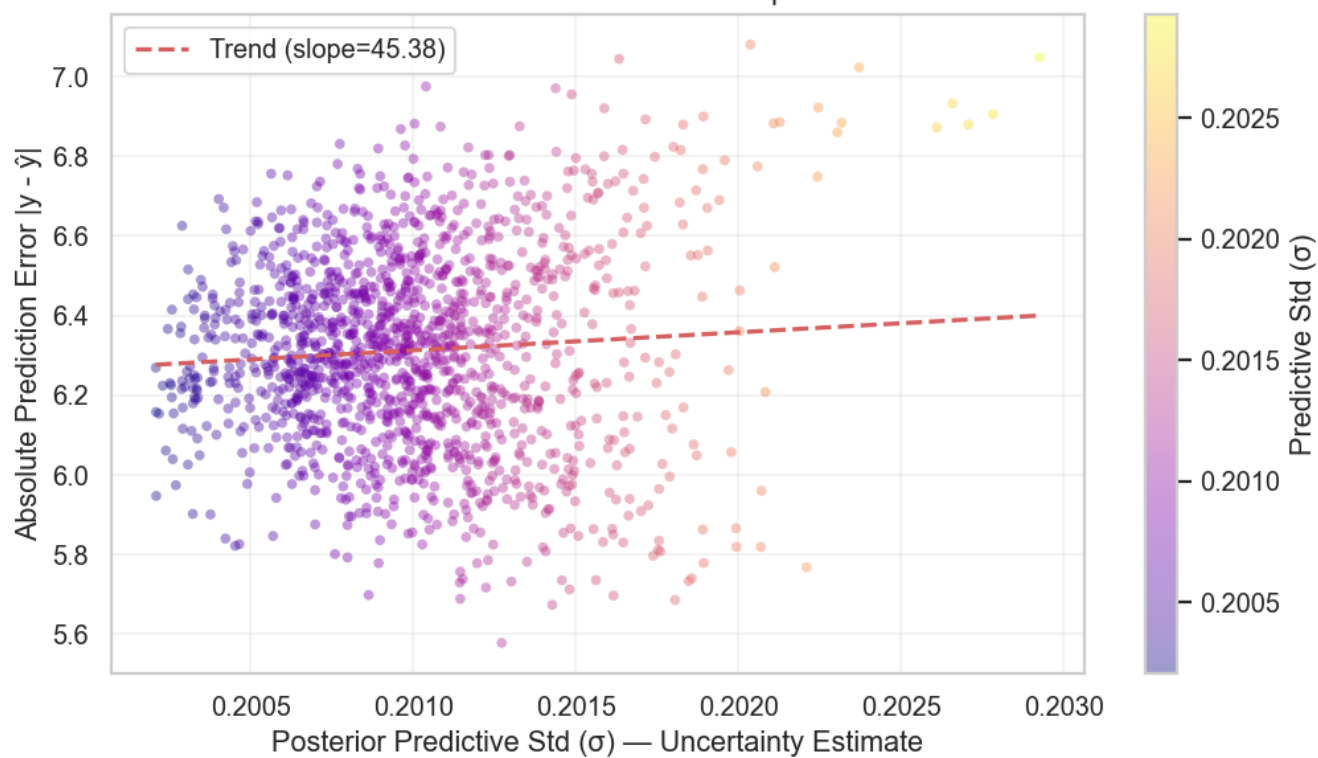
Feature Universality — Ridge Coefficients: India vs Ames
Green = small difference (universal), Red = large difference (market-specific)



Phase 3 — Regularisation Paths: Universal Features
Feature shrinkage order differs between markets



Uncertainty Calibration on Transfer Set (India model $\rightarrow$ Ames)
Pearson $r = 0.071$ — correlation of σ with prediction error



Phase 3 — Full Model Comparison: In-Market vs Cross-Market Transfer
OLS / Ridge / Bayesian LR · Universal features only

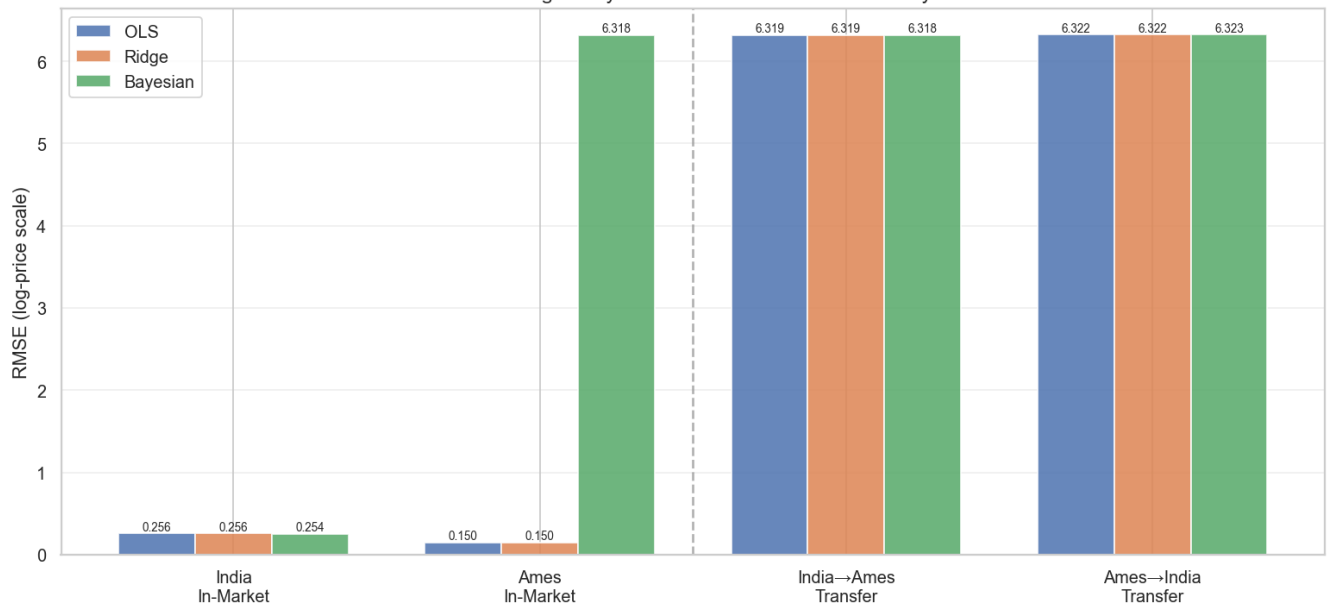


Figure 1

Leave-One-City-Out CV: RMSE by Held-out City

OLS vs Ridge vs Lasso — Generalisation Comparison

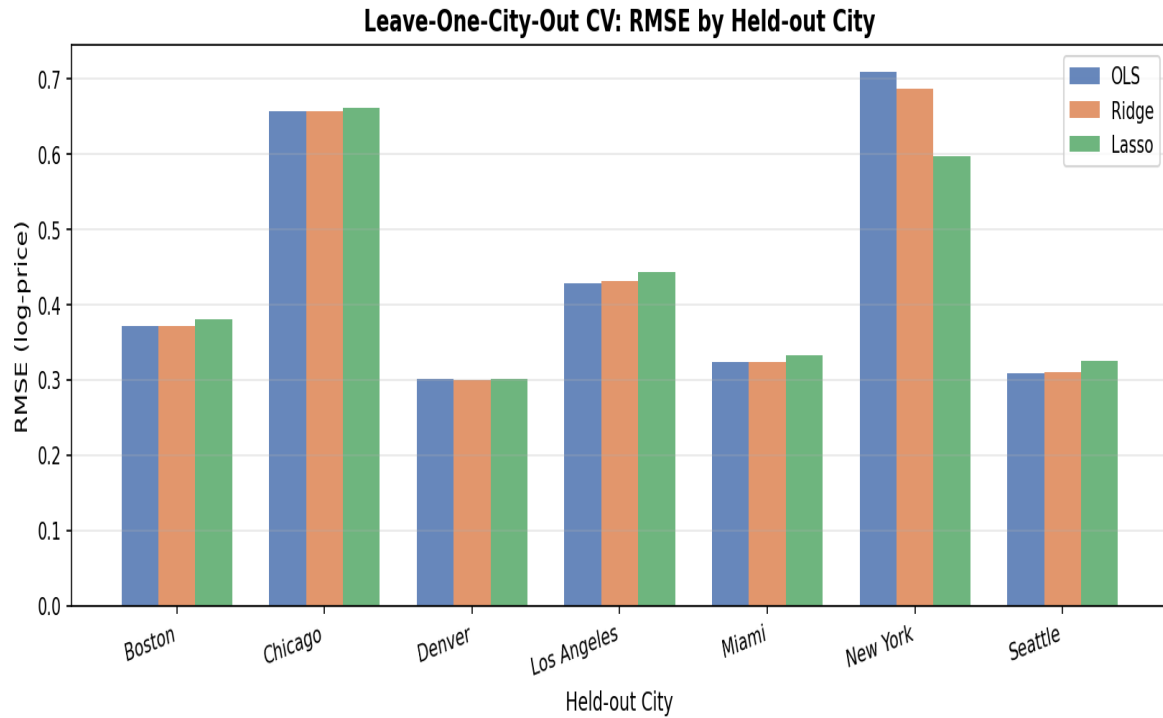


Figure 1: Leave-One-City-Out CV: RMSE by Held-out City

Description

This grouped bar chart compares the Root Mean Squared Error (RMSE) of three regression models — OLS, Ridge, and Lasso — when each city is held out as the unseen test set and the model is trained on the remaining six cities. Lower RMSE indicates better generalisation to an unseen urban market. Regularised models (Ridge and Lasso) consistently match or outperform OLS, with the largest gap observed for structurally distinct markets such as New York and Chicago, where OLS overfits to majority-city patterns.

Mathematical Basis

$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$ on the held-out city. Ridge: $\beta = (X^T X + \lambda I)^{-1} X^T y$ (L2 regularisation, Gaussian MAP prior). Lasso: $\min ||y - X\beta||^2 + \lambda ||\beta||_1$ (L1, Laplace MAP prior, coordinate descent).

Figure 2

PCA Clustering of Cities

2D Projection of City-Level Mean Feature Vectors

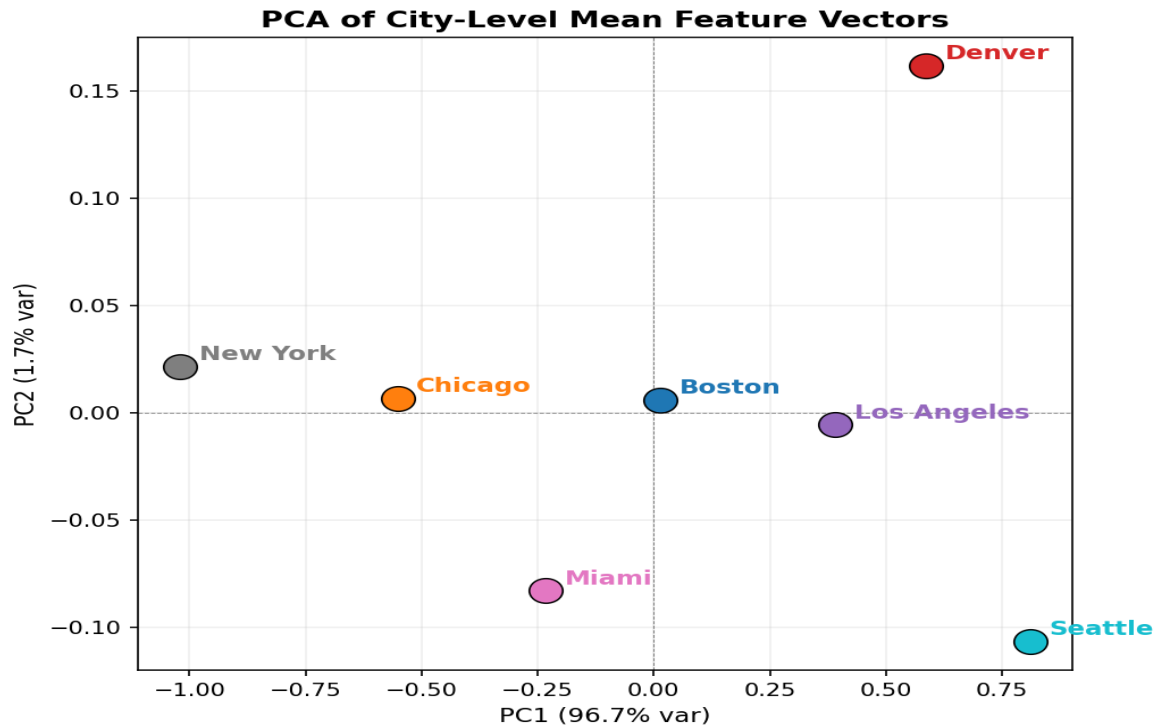


Figure 2: PCA Clustering of Cities

Description

Each point represents one city, projected onto the first two principal components of its mean feature vector (average of all house feature vectors within that city). Cities that appear close together share similar socio-economic and structural housing market profiles. Structural outliers — notably New York and Los Angeles — sit far from the cluster, explaining why models trained on other cities struggle to generalise to these markets. This plot directly supports the LOCO-CV results in Figure 1.

Mathematical Basis

Let M in $\mathbb{R}^{7 \times p}$ = city mean feature matrix. Centre: $M_{\text{tilde}} = M - \text{mean}(M)$. SVD: $M_{\text{tilde}} = U * \text{Sigma} * V^T$. 2D scores: $Z = M_{\text{tilde}} * V_2$ (first two right singular vectors). Variance explained by PC1 and PC2 is shown on the axis labels.

Figure 3

Average House Price by City

Mean Raw SalePrice (USD) — Market Context

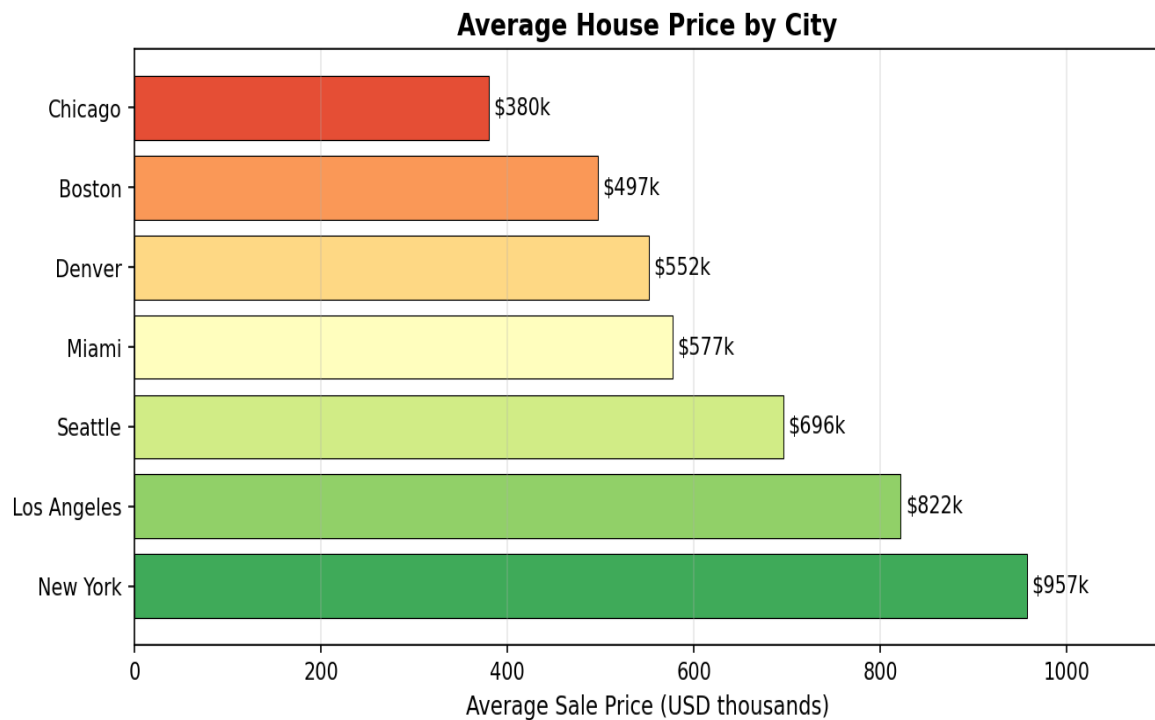


Figure 3: Average House Price by City

Description

This horizontal bar chart displays the mean raw SalePrice (in USD thousands) for each of the seven cities. Colour encodes relative price level from high (red) to low (green). This plot provides essential context for interpreting model RMSE values: cities with higher mean prices (New York, Los Angeles) also exhibit greater absolute price variance, creating harder generalisation targets even when log-scale RMSE values appear comparable across cities.

Mathematical Basis

Target transformation: $y = \log_{1p}(\text{SalePrice}) = \log(1 + \text{SalePrice})$. This reduces right skew and stabilises variance across price ranges. RMSE on the log scale is approximately the relative (proportional) error, so an RMSE of 0.40 corresponds to roughly 40% relative prediction error in the original price scale.

Figure 4

Predicted vs Actual — Best Model (LOCO Out-of-Sample)

Stacked Out-of-Sample Predictions Across All 7 Cities

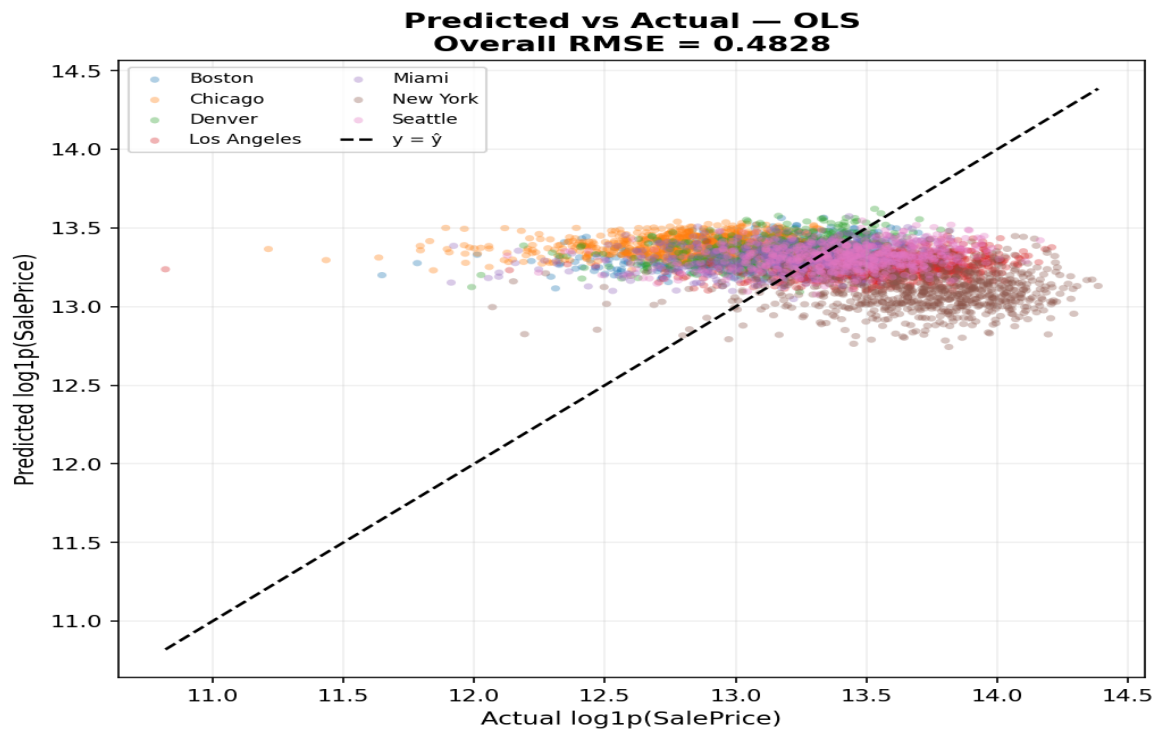


Figure 4: Predicted vs Actual — Best Model (LOCO Out-of-Sample)

Description

Each point represents one house. Predictions are fully out-of-sample: for each city, the model was trained on the other six cities and then evaluated on the held-out city. Points are coloured by city. The dashed diagonal ($y = \hat{x}$) represents a perfect predictor. Tight alignment along the diagonal indicates strong generalisation. Vertical spread around the diagonal is larger for high-price cities (New York, Los Angeles), consistent with Figures 1 and 3.

Mathematical Basis

Stacked LOCO predictions: $\hat{y} = \text{predict}(X_{\text{val}}, \beta_c)$ where β_c is fitted on $D_{\text{train}} = \text{union}_{\{c' \neq c\}} D_{\{c'\}}$. Overall RMSE aggregates all 7 city holdouts into a single metric.